Water Quality: Standards, Contaminants, and Treatment Technologies

Why Water Quality Matters and Regulatory Standards

Water quality defines the chemical, physical, and biological characteristics of water relative to its designated use. World Health Organization (WHO) Guidelines and national Environmental Protection Agency (EPA) regulations specify Maximum Contaminant Levels (MCLs) for physical parameters (pH, turbidity, total dissolved solids), chemical pollutants (nitrates, heavy metals, synthetic organics), and biological pathogens.

Water quality monitoring networks utilize continuous automated sensors, turbidity meters, pH probes, electrical conductivity (EC) sensors, dissolved oxygen (DO) meters, and gas chromatography-mass spectrometry (GC-MS) laboratory analysis to detect contamination early.

Sources of Pollution: Point and Non-Point Source Pollution

Point source pollution originates from identifiable single locations, such as municipal wastewater treatment plant outfalls, factory effluent pipes, and industrial facilities. Point sources are regulated via discharge permits and effluent limitations.

Non-point source (diffuse) pollution occurs when rainfall runoff washes land contaminants into waterways. Key non-point sources include agricultural fertilizers (nitrogen and phosphorus causing severe eutrophication and toxic microcystin algal blooms), urban stormwater runoff carrying heavy metals and microplastics, and sediment erosion.

Chemical Contaminants: Heavy Metals, Nitrates, and PFAS

Heavy metals such as lead, cadmium, mercury, and hexavalent chromium enter water through mining, industrial discharge, and corroding lead plumbing pipes. Chronic exposure leads to severe neurological damage, kidney failure, and developmental disorders.

Per- and polyfluoroalkyl substances (PFAS), known as 'forever chemicals', are persistent industrial compounds that accumulate in water systems and human blood. Nitrate contamination from intensive agriculture causes methemoglobinemia (blue baby syndrome) and requires advanced ion exchange or reverse osmosis treatment.

Water and Wastewater Treatment Processes

Conventional drinking water treatment follows a multi-stage process: coagulation/flocculation (adding alum to bind suspended solids), sedimentation (settling flocs), sand filtration, and chemical disinfection (chlorination, ozonation, or ultraviolet UV irradiation).

Wastewater treatment plants utilize primary physical settling, secondary biological activated sludge aeration tanks to decompose organic matter, tertiary nutrient removal (denitrification and phosphorus precipitation), and membrane bioreactors (MBR) or reverse osmosis (RO) for indirect potable reuse.